HDOC Day-67

Recursive GCD of Two Positive Integers Using Euclid's Algorithm

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Problem Statement

Write a recursive function to find the **GCD of two positive integers** using **Euclid's algorithm**.

Key Idea

Euclid's algorithm repeatedly replaces **GCD(a, b)** with **GCD(b, a % b)**. When **b becomes 0**, the answer is **a**.

Algorithm / Pseudocode

```
gcd(a, b)
    if b == 0
        return a
    else
        return gcd(b, a % b)
```

C Program

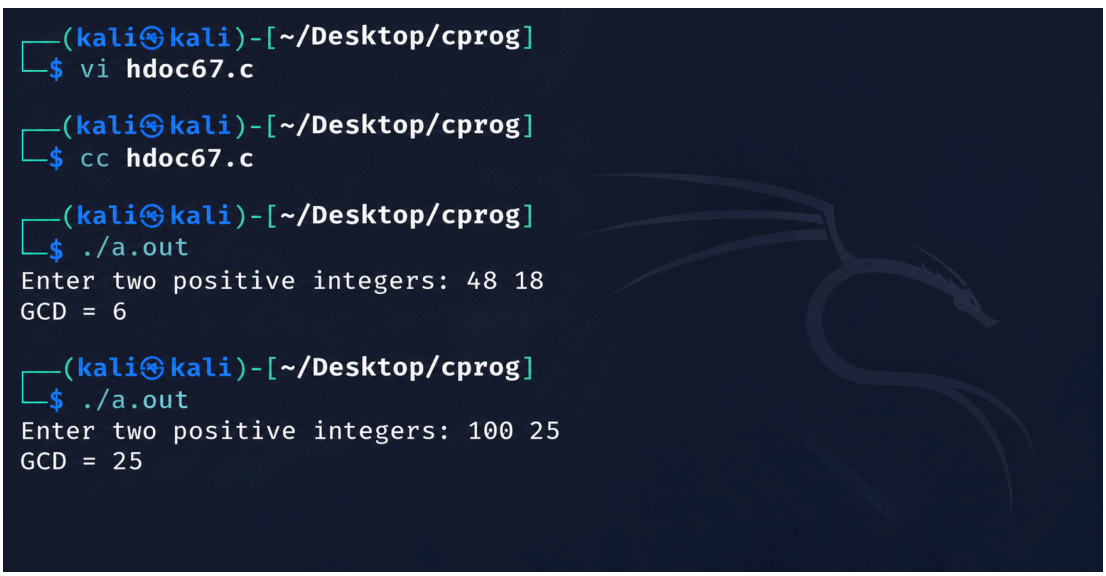
```c
#include <stdio.h>

int gcd(int a, int b)
{
    if (b == 0)
        return a;
    return gcd(b, a % b);
}

int main()
{
    int a, b;

    printf("Enter two positive integers: ");
    scanf("%d %d", &a, &b);
```

Kali Terminal-Style Program Run



Test Cases

Test Case	Input (a, b)	Expected GCD	Actual GCD
1	48, 18	6	6
2	100, 25	25	25
3	17, 13	1	1

Result

The recursive function successfully finds the GCD of two positive integers using Euclid's algorithm.